

SEAT No. _____

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[71]A-7

SARDAR PATEL UNIVERSITY

B.C.A. 3rd Semester (CBCS) (Reg.) Examination

Subject Code: US03BCA25

Subject Title: Data Structures - I

~~Saturday~~, 2-1-2021

Time: 2:00 P.M. To 4:00 P.M.

Max. Marks – 70

Q.1 Give Answers of Following Multiple Choice Questions. [10]

- I. An array is a _____ data structure.
(A) Composite (B) Non-composite
(C) Linear (D) Heterogeneous
- II. _____ is the basic fact or entity that is utilized in calculation.
(A) Data (B) Information
(C) Fact (D) All of these
- III. _____ is the building blocks of program.
(A) Information (B) Data
(C) Data Structure (D) Data Types
- IV. Which of the following is an operation of a Stack data structure?
(A) Push (B) Peep
(C) Pop (D) All of the above
- V. In the _____ notation the operators are written after the operands.
(A) prefix (B) postfix
(C) infix (D) None of these
- VI. Which of the following statement is FALSE for the Stack data structure?
(A) Its nature is LIFO (B) Its nature is FIFO
(C) It is a Linear data structure (D) It is a non-primitive data structure
- VII. A Queue is logically a _____ type of list.
(A) First-in First-out (B) First-in Last-out
(C) Last-in First-out (D) Last in Last-out
- VIII. Whenever an element is deleted from the Queue _____
(A) Front incremented by 1 (B) Front decremented by 1
(C) Rear incremented by 1 (D) Rear decremented by 1
- IX. Record is also known as group or _____.
(A) Item (B) Segment
(C) Entity (D) None of These
- X. A record item that uniquely identifies a record in a file is called a _____.
(A) Key (B) Sequence Key
(C) Item (D) None of these

[P.T.O.]

[1]

Q.2 Fill in The Blanks And True-False.**[08]**

- I. Index of an array is always an _____ value.
- II. When new data are to be inserted into a data structure, but there is no available space this situation is usually called _____
- III. The line formed for paying fees for college is an example of _____
- IV. A digraph is known as _____ graph.
- V. Type of the array represents the kind of data type. **[True/False]**
- VI. An operation that is used to insert an element on a stack is known as Pop. **[True/False]**
- VII. A circular queue overcomes the problem of unutilized space in linear queues implemented as arrays. **[True/False]**
- VIII. The collection of files is known as Datafile. **[True/False]**

Q.3 Answer the Following Short Questions. (Attempt Any Ten)**[20]**

- I. What is Linear Data structure? List out Linear data structure.
- II. List out advantages of Data Structure.
- III. List out applications of an array.
- IV. What is Push and Pop operations of a Stack?
- V. What do you mean by Top and Bottom of a Stack?
- VI. List types of notation.
- VII. State the applications of Queue.
- VIII. Define queue. List the types of Queues.
- IX. What are Descending Priority Queues? Explain in brief.
- X. Define Following Terms (I). **Graph** (II). **Weighted graph**.
- XI. What do you mean by Transaction? Which types of transactions are performed on the file?
- XII. Which criteria are used for selection of file organization?

Q.4 Answer the Following Long Questions. [Attempt Any Four]**[32]**

- I. Explain in brief Classification of Data Structure.
- II. List out operations used on Data Structure. Explain any three.
- III. Write an algorithm to insert an element into a Stack.
- IV. Explain Infix to Postfix conversion.
- V. Explain the Simple Queues with an example. Also explain the limitations of Simple Queues.
- VI. Explain the Circular Queues with an example.
- VII. Write a short note on Multiple buffering.
- VIII. Explain in detail the structure of Sequential File.